

Quadratic form:

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. The following expression

$$\boxed{q(x) = x^T A x}$$

is called a quadratic form

where x is a n -D vector.

Ex:

$$A = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix}_{3 \times 3}$$

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{aligned} q(x) &= x^T A x = (x_1 \ x_2 \ x_3) \begin{pmatrix} 1 & -1 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\ &= x_1^2 + 3x_2^2 + 4x_3^2 - 2x_1x_2 + 4x_1x_3 + 2x_2x_3 \end{aligned}$$

$$q(x) = x_1^2 - 2x_2^2 - 4x_1x_3, \quad A = ?$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

Positive - semi definite (PSD) matrices:

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is said to be PSD $\iff$ for all $x \in \mathbb{R}^n$
 we have $q(x) = x^T A x \geq 0$

Ex:

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 10 \end{pmatrix}_{2 \times 2} \checkmark$$

$$x^T = (x_1, x_2)$$

$$\begin{aligned} x^T A x &= x_1^2 + 10x_2^2 + 6x_1x_2 \checkmark \\ &= (x_1 + 3x_2)^2 + x_2^2 \geq 0 \end{aligned}$$

$\therefore A$ is a PSD matrix

Positive Definite (PD)

$$q(x) = x^T A x > 0 \quad \begin{matrix} \nearrow \text{strictly} \\ \nearrow \text{strictly} \end{matrix} \text{ then}$$

- * PSD $\rightarrow$ Eigen values are non-negative (≥ 0)
- * PD $\rightarrow$ " " +ve

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\det = 2 > 0$$

$$\det = 3 > 0$$

$$\det = 4 > 0$$

PD

Inner Product

$\vec{u}$

$\vec{v}$

dot product

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$(\vec{u}^T \vec{v})$$

$\hookrightarrow$ scalar

Outer Product

$\hookrightarrow$ o/p is a matrix

$$u v^T$$

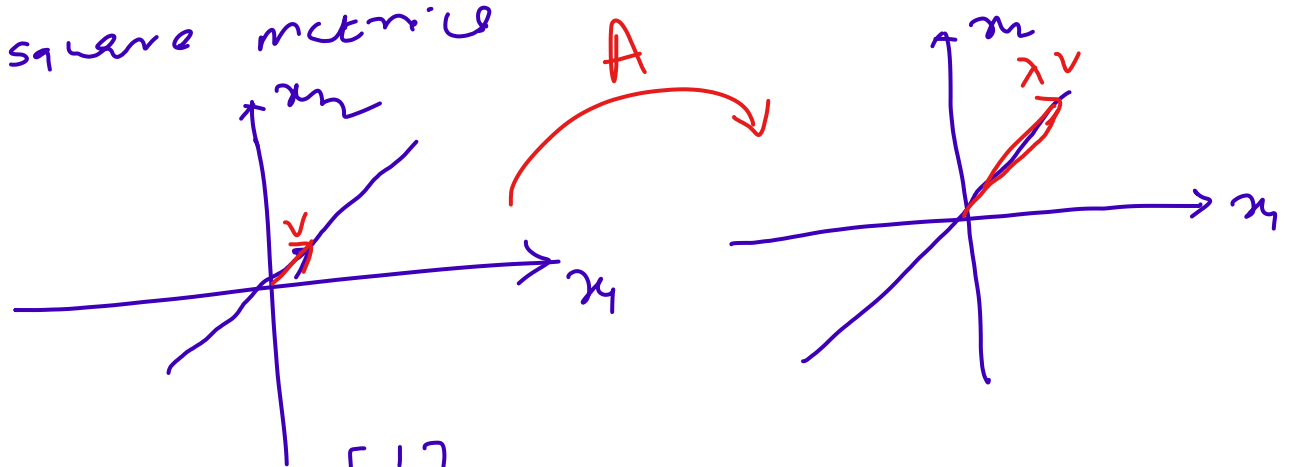
$$(3 \times 1)(1 \times 3)$$

$$\times \hookrightarrow 3 \times 3$$

Eigen Decomposition:

Defined for square matrices

$$\boxed{Av = \lambda v}$$



$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

$$\lambda_1 = 3, v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1, v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}_A \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{v_1} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{\lambda_1 v_1} \rightarrow \textcircled{1}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}_A \begin{bmatrix} -1 \\ 1 \end{bmatrix}_{v_2} = -1 \begin{bmatrix} -1 \\ 1 \end{bmatrix}_{\lambda_2 v_2} \rightarrow \textcircled{2}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}_A \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}_{v_1 v_2} = \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix}_{\lambda_1 v_1 \lambda_2 v_2}$$

$$\begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}_{v_1 v_2} \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}_{\lambda_1 \lambda_2}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

A
 $\underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_V$
 $\underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_V$
 $\underbrace{\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}}_\Lambda$

$$\Rightarrow \boxed{AV = V\Lambda}$$

$A = ?$

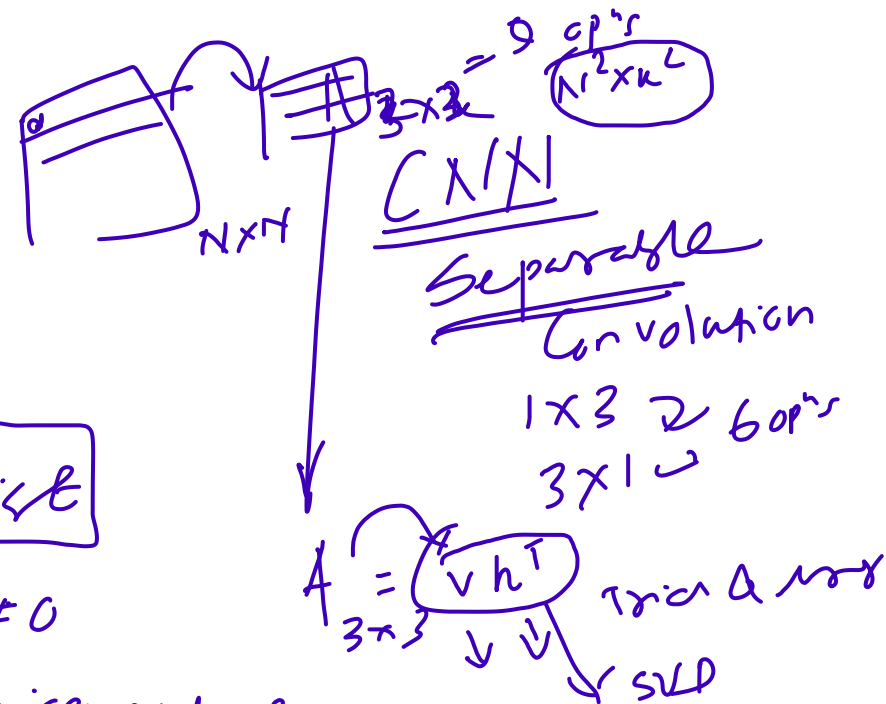
$$AVV^{-1} = V\Lambda V^{-1}$$

$$\Rightarrow A I = V\Lambda V^{-1}$$

$$\Rightarrow \boxed{A = V\Lambda V^{-1}}$$

$$\boxed{A}_{n \times n} = \underbrace{\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}}_V \underbrace{\begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}}_\Lambda \underbrace{\begin{bmatrix} | & | & & | \\ & & & \\ & & & \\ & & & \end{bmatrix}}_{V^{-1}}_{n \times n}$$

$\lambda_1, \lambda_2, \dots, \lambda_n$ distinct eigen values
 $v_1, v_2, \dots, v_n$ eigenvectors
 V^{-1} inv of V



$$\boxed{\lambda_1, \lambda_2, \dots, \lambda_n}$$

Limitations: ED can only handle square matrix.

Necessary Conditions: for $n \times n$ matrix, we should have n 'LI' eigenvectors.

A is a ^{square} symmetric matrix

$\left\{ \begin{array}{l} \text{eigen values} \rightarrow \text{real \& +ve} \\ \text{eigenvectors} \rightarrow \text{orthogonal} \end{array} \right.$

$$A = V \Lambda V^{-1}$$
$$\Rightarrow \boxed{A = V \Lambda V^T}$$

$V \Rightarrow$ Orthogonal matrix
 $V^{-1} = V^T$

Singular Value Decomposition (SVD):

$$A = U \Sigma V^T$$

A is $m \times n$
 U is $m \times m$ square
 Σ is $m \times n$ diagonal matrix
 V^T is $n \times n$ square
 $m \neq n$

$$U = \begin{bmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_m \\ | & | & \dots & | \end{bmatrix}_{m \times m}$$

$u_1, \dots, u_m$ are $(m \times 1)$ orthonormal vectors
 left singular vectors

$$V = \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{bmatrix}$$

$v_1, v_2, \dots, v_n$ are $(n \times 1)$ orthonormal vectors
 right singular vectors

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & \dots & & \\ & & \ddots & & & \\ & & & \sigma_m & & \\ 0 & 0 & \dots & & 0 & 0 \\ & & & & \ddots & \\ 0 & 0 & \dots & & & 0 \end{bmatrix}_{m \times n}$$

$\sigma_1, \sigma_2, \dots, \sigma_m$ are singular values
 (positive)

$$\sigma_1 > \sigma_2 > \dots > \sigma_m$$

V & Σ :

$$A = U \Sigma V^T$$

$$\begin{aligned} A^T A &= (\underbrace{U \Sigma V^T}_A)^T \underbrace{U \Sigma V^T}_B = (\Sigma V^T)^T U^T U \Sigma V^T \\ &= V \Sigma^T \underbrace{U^T U}_I \Sigma V^T \\ &= V \Sigma^T \Sigma V^T \\ &\approx V \Sigma^2 V^T \end{aligned}$$

$A^T A \rightarrow$ Square symmetric matrix

$n \times n$

$$(\underbrace{A^T A}_B)^T = \frac{A^T A}{D}$$

$$\underbrace{A^T A} = V \underbrace{\Sigma^T \Sigma}_V V^T$$

$\downarrow$
 $V^{-1} = V^T$

exactly n eigenvalues
 n mutually orthogonal vectors

$$\begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & & \\ & & \ddots & \\ 0 & & & \sigma_n^2 \end{bmatrix}$$

$$\boxed{A^T A = V \Sigma^T \Sigma V^{-1}}$$

$$= A = V \Lambda V^{-1}$$



orthogonal
vectors

Eigen vectors of $A^T A = v_1 v_2 \dots v_r$

$$A = U \Sigma V^T$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad \Sigma^2 = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$\sigma_1^2 = \lambda_1 \Rightarrow \sigma_1 = \sqrt{\lambda_1}$$

$$\sigma_2 = \sqrt{\lambda_2}$$

$$\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 & \dots \\ 0 & \sqrt{\lambda_2} & \dots \\ 0 & \dots & \sqrt{\lambda_m} \\ 0 & \dots & 0 \\ 0 & \dots & 0 \end{bmatrix}_{m \times n}$$

$$\begin{matrix} 2 \times 2 \\ \lambda_1 & \lambda_2 \end{matrix}$$

$$4 \times 2$$

$$AA^T = U \Sigma \Sigma^T U^T \approx U \Sigma^2 U^T$$

m eigen values

non-zero eigen values are same

$$m > n$$

$$m < n$$

4×4

$$A = U \Sigma V^T$$

$\swarrow$ eigen vectors of AA^T
 $\searrow$ Σ $\swarrow$ $AT A$ $\rightarrow$ sq. root of eigen values in the principal diagonal elements

Reduced SVD:

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}_{3 \times 2}$$

$$V \xrightarrow{U, A} A^T A$$

$$V = \begin{bmatrix} \checkmark \\ \downarrow v_1 \\ \downarrow v_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}_{2 \times 2}$$

$$\checkmark \Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A = U \Sigma V^T$$

$\downarrow$
 3×3 3×2 2×2

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$$

$$\lambda_1 = 3$$

$$\lambda_2 = 1$$

$$V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2 \times 1} \xrightarrow{\sqrt{2}} \bar{V}_1$$

$$V_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}_{2 \times 1} \xrightarrow{\sqrt{2}}$$

$$\xrightarrow{\approx A} U \Sigma V^T \rightarrow$$

3×1 1×1 1×2

$$3 \times 2$$

$U \xrightarrow{\text{eVs } 4 \text{ } AA^T}$

$$AA^T = \begin{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} \end{bmatrix}_{3 \times 3}$$

$$\begin{aligned} \lambda_1 &= 3 \\ \lambda_2 &= 1 \\ \lambda_3 &= 0 \end{aligned}$$

$$\begin{aligned} u_1 &\rightarrow 3 \times 1 \\ u_2 &\rightarrow 1 \times 1 \\ u_3 &\rightarrow 1 \times 1 \end{aligned}$$

$$\begin{bmatrix} u_1 & u_2 & u_3 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

②

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -9 \end{bmatrix}_{2 \times 3}$$

$$U \Sigma V^T$$

$2 \times 2 \quad 2 \times 3 \quad 3 \times 3$

$$A^T A = \begin{matrix} \lambda_1 & \lambda_2 & \lambda_3 \\ \sqrt{\lambda_1} & \sqrt{\lambda_2} & \sqrt{\lambda_3} \end{matrix}$$

$$\begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \sqrt{\lambda_2} & 0 \end{bmatrix}$$

$$AA^T_{2 \times 2}$$

$$\begin{aligned} \lambda_1 &= u_1 \\ \lambda_2 &= u_2 \end{aligned}$$

Full SVD:

$(m > n)$

$A_{m \times n}$

$$A_{m \times n} = U_{m \times m} \Sigma_{m \times n} V_{n \times n}^T$$

$u_i \rightarrow m \times 1$
 $v_i \rightarrow n \times 1$

$$\begin{bmatrix} \sigma_1 & 0 & 0 & \dots \\ 0 & \sigma_2 & & \\ 0 & & \dots & \sigma_n \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

$(m < n)$

$$\begin{bmatrix} \sigma_1 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & & & \\ 0 & \dots & \sigma_m & 0 & \dots \end{bmatrix}$$

Reduced SVD:

$m > n$

$$A =$$

$m \times n$

$$U$$

$m \times m$

$$\Sigma$$

$n \times n$

$$V^T$$

$n \times n$

4×2

$$A = U \Sigma V^T$$

$m < n$

2×4

$$= U \Sigma V^T$$

$\sigma_1 > \sigma_2 > \dots > \sigma_n$

$$\begin{bmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_m \\ | & | & \dots & | \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & 0 & \dots \\ 0 & \sigma_2 & 0 & \dots \\ 0 & \dots & \dots & \sigma_n \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{bmatrix}$$

4×4 $m \times n$ 4×2

4×2 2×2

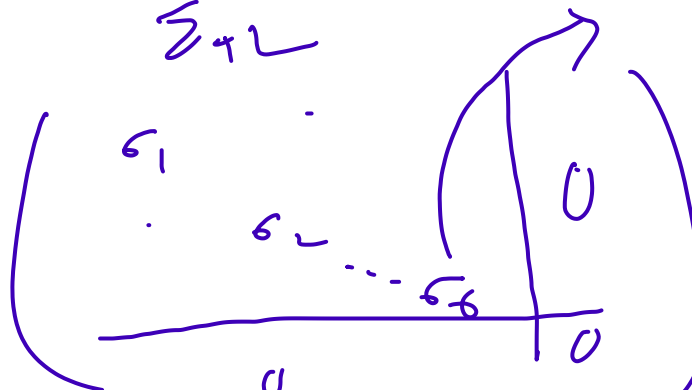
$$V = n \times m$$

$$\begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{bmatrix}$$

4×1

$$V = 4 \times 2$$

$$A = U \Sigma V^T$$

$$= \begin{pmatrix} u_1 & u_2 & u_3 & \dots & u_m \\ U_L & U_R \end{pmatrix} \begin{pmatrix} \Sigma_{TL} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 & v_2 & \dots & v_r & \dots & v_n \\ V_L & V_R \end{pmatrix}$$


$$= (U_L | U_R) \begin{pmatrix} \Sigma_{TL} & 0 \\ 0 & 0 \end{pmatrix} (V_L | V_R)^T$$

$$= (U_L \Sigma_{TL} | 0) \begin{pmatrix} V_L^T \\ V_R^T \end{pmatrix}$$

$$= U_L \Sigma_{TL} V_L^T$$

$$= (u_1 \ u_2 \ \dots \ u_r) \begin{pmatrix} \sigma_1 & \sigma_2 & \dots & \sigma_r \\ & & & \end{pmatrix} (v_1 \ v_2 \ \dots \ v_r)^T$$

$$A = U \Sigma V^T$$

$m \times n$ $m \times m$ $m \times n$ $n \times n$

k singular values,

$$A_{m \times n} =$$

$$U_{m \times k} \Sigma_{k \times k} V^T_{k \times n}$$

$$V \rightarrow n \times k$$

CNN

Convolution:

Separable Convolution:

$2 \times n \times k$

1×3

3×1



$$K = v h^T$$

3×3

$\rightarrow$ SVD

$= k$

3×3

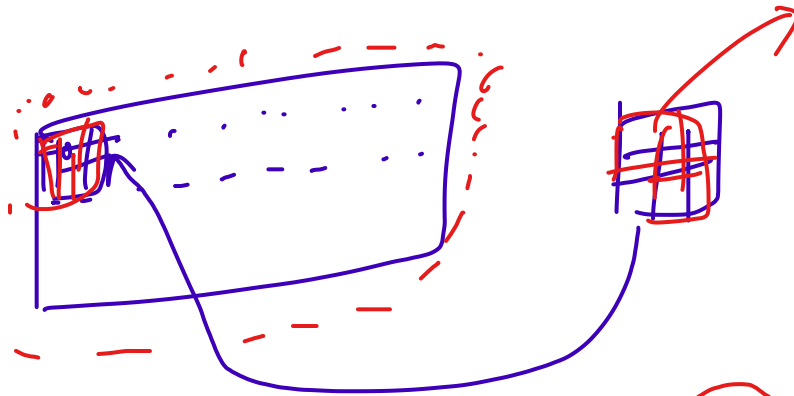
9 parameters

5×5 kernel

25 parameters

$3 \times 3 \rightarrow 9$

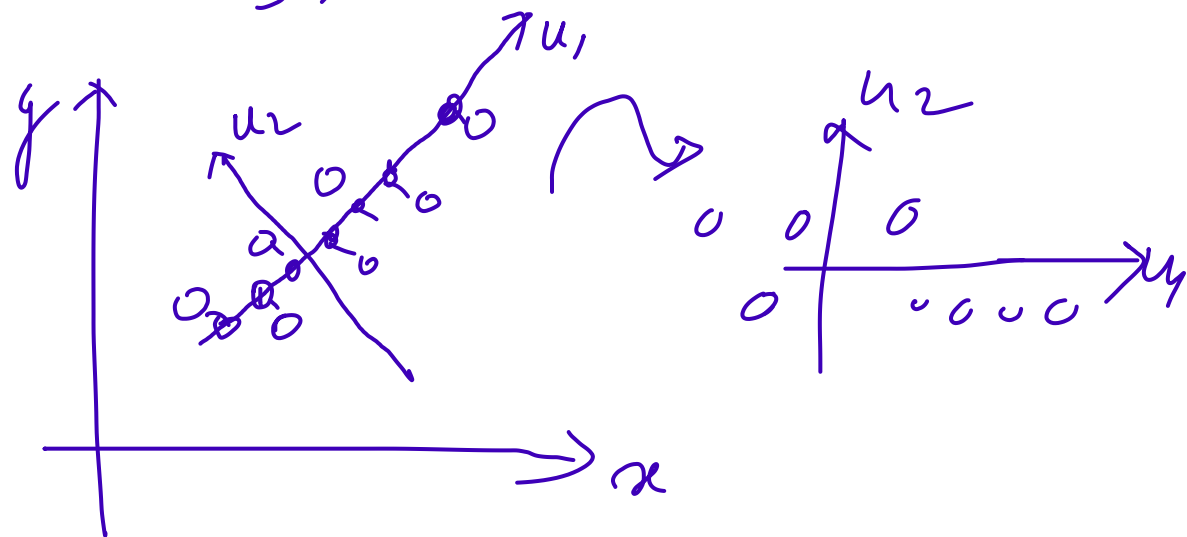
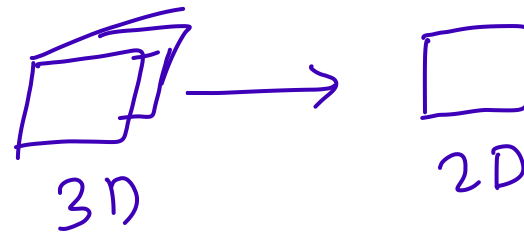
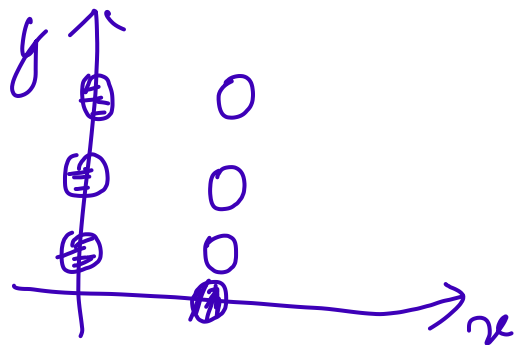
$17 \times 17 \rightarrow 14$



PCA :

$$\begin{matrix} N \times d \\ \underline{\quad} \end{matrix} \xrightarrow{P^T A} \begin{matrix} N \times d' \\ \underline{\quad} \end{matrix}$$

$d' \ll d$



Project
high dimensional data

↓
New coordinate system

①

②

Given
 $D: X_{n \times d}$

Project the data X on a vector $\vec{u}$ s.t.

- ① The variance of the projected data is maximum
- ② u is an unit vector

Project $\vec{x}_i$ on $\vec{u}$

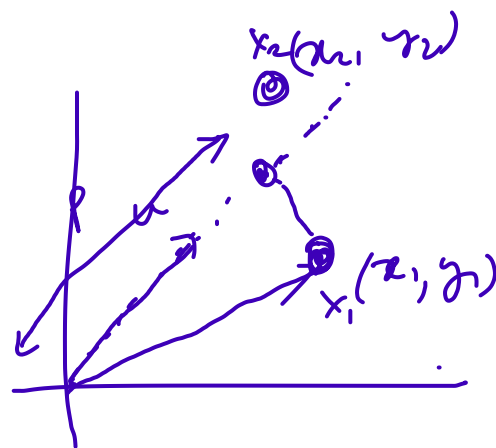
$$p = \frac{\vec{x}_i \cdot \vec{u}}{\|\vec{u}\|^2} = \vec{x}_i \cdot \vec{u}$$

if $u = (m, n)$, $= mx + ny$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times d}$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix}_{d \times 1}$$

$$\text{Proj}_u X = Xu = \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix}_{n \times 1}$$



Optimization:

$$\arg \max_u \underbrace{\text{Var}(Xu)}_{\rightarrow u^T \Sigma u}$$

$$\text{s.t. } \|u\| = 1$$

$$u^T \Sigma u$$

$\hookrightarrow$ Cov. matrix

$$\arg \max_u [u^T \Sigma u]$$

$$\text{s.t. } \|u\| = 1$$

Lagrange Multiplier

$$Av = \lambda v$$

$N \times d$
 $\nearrow$
 $\downarrow$

Sol:

$$\boxed{\Sigma u = \lambda u}$$

Variance $\rightarrow \lambda_1, \dots, \lambda_d$
 $u_1, \dots, u_d \rightarrow d \times 1$

PCs

$d \times d$

$$\Sigma = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

$\hookrightarrow d \times d$

Centered data

$$Y = X_c U$$

$n \times k$ $n \times q$ $d \times k$

$k = d'$

$$U = \begin{bmatrix} u_1 & u_2 & \dots & u_q \\ \vdots & \vdots & & \vdots \end{bmatrix}_{d \times d}$$

(k)

How to choose u ?

$$PC_k = \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^q \lambda_i}$$

$$\vec{u}_1 = \frac{\lambda_1}{\lambda_1 + \dots + \lambda_q}$$

$$\vec{u}_2 \dots \vec{u}_q = \frac{\lambda_1 + \lambda_2}{\lambda_1 + \dots + \lambda_q} \dots \text{There are } q \text{ of them}$$

(2) Steps: PCA $(X_{n \times q}, k)$

(1) Compute cov. matrix $\Sigma_{d \times d}$

(2) Compute eigen values & eigen vectors $\lambda_1 \dots \lambda_q$

(3) Sort the eigen values acc. to eigen values

(4) Partition U matrix

(5) $Y = X_c U$